

Raghav Gupta

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Summary

Machine Learning Scientist with 4+ years experience across deep learning research and production ML. Working on multimodal deep learning, vision-language-action models, VLM mid-training/fine-tuning, diffusion generation and computer use agent, with strong foundational skills and proven production impact.

Education

Carnegie Mellon University

M.S. in Machine Learning and Data Science, GPA: 4.0/4.0

Pittsburgh, PA

Aug 2025 – Dec 2026

Coursework: Generative AI, Deep Learning Systems, Cloud Computing, Multimodal Machine Learning

Indraprastha Institute of Information Technology Delhi

Delhi, India

B.Tech. in Computer Science, GPA: 9.21/10.00

Aug 2017 – Jun 2021

Coursework: Probability and Statistics, Deep Learning, Computer Vision, Natural Language Processing

Technical Skills

Languages: Python, SQL, C++, CUDA

Frameworks/Tools: PyTorch, Scikit-learn, NumPy, Pandas, BigQuery, Vertex AI, GCP, Git, Weights & Biases

Core Areas: Deep Learning, Generative AI, Multimodal Learning, Time-Series Forecasting, Parameter-Efficient Fine-Tuning

Current Research

VLAFactory: Data-Centric Pipeline for VLA Model Training [\[code\]](#)

EMNLP 2026 Demo (submitted)

- Wrote and ran the **VLM mid-training** code for this open-source **Vision-Language-Action (VLA)** pipeline, training **InternVL2/InternVL3.5** backbones with **full fine-tuning, LoRA, and QLoRA** across six data recipes (LAION re-captioning, nuScenes scene captions, DriveLM VQA).
- Ran the downstream **autonomous-driving** experiments over general (MMMU, MME, SEED) and embodied (VLADBench, RealWorldQA) VLM benchmarks and closed-loop driving: mid-training lifted **VLADBench 16.9→40.9** and **Bench2Drive score 83.3→86.6** (success **60.9%→68.1%**), with intermediate VLM gains transferring inconsistently downstream.

Experience

Fast Retailing (UNIQLO)

Tokyo, Japan

Machine Learning Scientist

Aug 2021 – Jun 2025

- Deployed large-scale demand forecasting systems on Vertex AI using hierarchical forecasting, probabilistic modeling, and feature-rich signals including lags, seasonality, holidays, and embedding-based item similarity; reduced **WAPE by 3 points** and cut excess inventory by **¥6B (\$39M)**.
- Automated product image classification and metadata generation using **multimodal LLMs**, eliminating **100+ hours** of manual labeling and improving downstream analytics workflows.
- Engineered **20TB+** BigQuery feature pipelines powering SKU-level forecasting models: rolling-window aggregations, moving averages, normalization, and threshold-based transforms across **19 product categories**.

Selected Projects

GoT-Diffuse: Compositional Text-to-Image Generation [\[report\]](#)

SDXL, Diffusion, PyTorch

- Identified a **planner-decoder execution gap** in reasoning-based **text-to-image** generation on **T2I-CompBench++**. Plans are spatially valid **83.1%** of the time but only **64.7%** of images satisfy the relations, pinpointing compositional *binding*, not object rendering, as the bottleneck via a custom BLIP-VQA/UniDet diagnostic and a five-metric text-encoder suite.
- Showed **layout-guided SDXL** conditioning lifts spatial relation preservation **+19%** (0.17→0.21) and color binding **+15%** (0.42→0.48) over baseline SDXL, and that plan corrections help only when kept in-distribution, motivating **GoT-Diffuse**, a timestep-routed diffusion decoder that preserves object-specific conditioning during early denoising.

TS-QFormer: Time-Series-to-LLM Adapter for QA [\[report\]](#)

PyTorch, Frozen Encoders/LLMs

- Built a parameter-efficient adapter connecting a frozen **MOMENT-large** time-series encoder to a frozen **OPT-2.7B** language model using channel-aware projections and question-conditioned query tokens.
- Developed a supervised chain-of-thought training pipeline for temporal reasoning over time-series inputs, improving ROUGE-F1 over a direct-answer baseline.

Deep Learning Stack from Scratch

Python, C++, CUDA

- Implemented reverse-mode autograd, stable softmax loss, SGD/Adam optimizers, and core neural network modules including Linear, ReLU, normalization, and dropout.
- Built a NumPy-like NDArray backend with stride-aware tensor operations and tiled GPU matrix multiplication kernels; added FlashAttention-style optimization reducing memory usage by **20%+** relative to vanilla attention.

Generative Modeling and Efficient Transformer Architectures

PyTorch

- Implemented **DDPM** with a U-Net backbone, ELBO objective, and cosine noise schedule, trained on **AFHQ-Cat**.
- Extended **minGPT** with **RoPE** and **Grouped-Query Attention**, improving validation loss by **10%+** while benchmarking throughput and memory tradeoffs.